

# Aryan Mehesare

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AI/ML & DevOps Engineer

## Summary

Pre-final year Computer Science undergraduate building AI-driven systems and full-stack applications. Experienced in developing and deploying machine learning pipelines with explainable models using PyTorch. Built end-to-end projects in audio deepfake detection and image super-resolution, integrating ML models into production-ready platforms.

## Education

### Vellore Institute of Technology

B.Tech in Computer Science and Engineering — CGPA: 8.81

Chennai, India

August 2023 – August 2027 (Expected)

### The Bishop's Co-Ed School

Grade 12: 82.2% | Grade 10: 98.4%

Pune, India

July 2009 – May 2023

## Experience & Leadership

### SpeedTech.ai

May 2026 – Present

Generative AI Intern

- Designed and managed multimodal MCP configurations across OpenAI and Gemini ecosystems for audio workflows.
- Optimized telephony-based AI voice agents, reducing response latency by 30–40% and improving performance.
- Architected multi-agent AI system with 6+ agents & an integrated knowledge base, automating CRM operations & workflows.
- Implemented AI-driven lead qualification, tracking, and follow-up automation, improving efficiency by 4×.

### Google Developer Groups (GDG), VIT Chennai

Oct 2024 – Mar 2026

Core Team Member

- Conducted workshops on Next.js and modern web architecture for 100+ student developers.
- Organized hackathons and technical events, providing mentorship and operational support to participants.

## Projects

### Audio Deepfake Forensics

Dec 2025 – Feb 2026

- Engineered a dual-feature **ResNet-18** & **CNN** ensemble, achieving **3.31% EER** and **0.9940 ROC-AUC** on the ASVspoof dataset.
- Applied **Grad-CAM** & **Integrated Gradients** to visualize vocoder artifacts, validating defense against 13 unseen zero-day attacks.
- Developed a **Gradio** dashboard delivering real-time multimodal forensic analysis with **<22 ms** inference latency.
- Optimized generalization using **SpecAugment** masking and 80:20 score-level fusion, scaling detection precision to **99.98%**.
- Tech Stack:** PyTorch, Captum (XAI), Librosa, Gradio, Scikit-learn

### Single Image Super Resolution (SRCNN)

Aug 2025 – Sep 2025

- Developed and trained a 3-layer SRCNN in PyTorch on **800+ high-res images** (DIV2K), achieving strong PSNR/SSIM scores.
- Compared model performance across 2×, 3×, and 4× scaling factors using **MLflow** with **50+ experiment runs**.
- Managed training infrastructure on **AWS SageMaker** with S3-backed data pipelines for **20GB+** of training data.
- Tech Stack:** PyTorch, MLflow, AWS SageMaker, S3, Python

### SafeScape – IoT & ML Safety Platform

Sep 2024 – Nov 2024

- Engineered a real-time safety platform with geofencing and safe routing algorithms using **Next.js** and **FastAPI**.
- Built an **OpenCV**-based person detection model using CCTV data and an audio classification model for detecting distress signals.
- Deployed containerized microservices on **AWS EC2** via Docker and Nginx as a reverse proxy.
- Tech Stack:** Next.js, FastAPI, OpenCV, PyTorch, PostgreSQL, Docker, AWS EC2

## Technical Skills

**Programming Languages:** Python, C/C++, JavaScript, TypeScript, Java, SQL, Shell

**AI/ML & Data:** PyTorch, scikit-learn, OpenCV, MLflow, NumPy, Pandas, Matplotlib, LangGraph

**Web & Backend:** Next.js, React.js, FastAPI, Node.js, PostgreSQL, MongoDB, Redis

**Cloud & DevOps:** AWS (EC2, S3, SageMaker, Lambda), Docker, Git, Linux, Nginx

## Certifications

- Oracle Cloud Infrastructure 2025 Generative AI Professional (1ZO-1127-25)
- AWS Certified Solutions Architect Associate 2026 — Udemy

## Achievements

- Winner** — Sanrakshan Hackathon, IIT Bombay Techfest 2024
- Top 10** — Smart India Hackathon (SIH) Internal Rounds (2023, 2024)